Inference-Time Scaling: How Much Does Extra Compute Actually Buy You?

The claim

Increasing inference compute improves accuracy up to a task-dependent saturation point, beyond which extra compute adds cost and latency without reliable accuracy gain. This is falsifiable: if accuracy rose without ever flattening — or never rose at all — the claim would fail. We test it directly across four distinct mechanisms for spending extra compute, on real models, with real data.

Why now

Reasoning models (o1, DeepSeek-R1, QwQ) made "think longer" a standard lever, but *how* that compute is spent varies enormously and is rarely compared head-to-head. Snell et al. (2024) formalized best-of-N and beam search as compute-optimal search strategies. Geiping et al. (2025) introduced recurrent-depth models (Huginn-0125) that scale compute by iterating a shared block in latent space rather than emitting more tokens. Muennighoff et al. (2025, *s1*) showed a third, almost trivially simple lever — budget forcing, suppressing a model's stop token and appending "Wait" to force continued reasoning. Pathway's Dragon Hatchling family (BDH, BDH-CQ) adds a fourth angle: reasoning entirely in undecoded latent space, with no verbal chain of thought at all.

Mechanism and comparison

Method	How compute is spent	Cost shape	Observability
Best-of-N	N independent samples, majority vote	Linear in N, parallelizable	Full — every sample is real text
Beam search	PRM scores each step, prunes, branches	Linear in beam width x depth	Full, but score-dependent
Latent compute (Huginn)	Same recurrent block applied r times before decoding	Linear in r, sequential per token	None — no verbal trace, only the r dial
Budget forcing	Suppress stop, inject "Wait," force continuation	Linear in forced continuations	Full — forced text is real reasoning
BDH-CQ dual recurrence	Memory (`S_t`) + reasoning (`H_r`) recurrence, discrete effort levels	Discrete (LOW/MED/HIGH), trained-in	None — `H_r` never decoded into language

We ran best-of-N and budget forcing live on Qwen2.5-1.5B-Instruct, and latent compute scaling live on Huginn-0125 (4-bit quantized to fit the full r ∈ {1,4,8,16,32,64} range in 8GB VRAM — this project's own quantization work, not published). All three are real, streamed, per-problem-checkable against GSM8K/OpenBookQA gold answers. Beam search and BDH-CQ's numbers are published, not reproduced by us, and labeled accordingly.

Where BDH and BDH-CQ fit

BDH-CQ is the primary connection: its Table 5 shows accuracy rising with effort level (LOW 21% → MEDIUM 27% → HIGH 29.5% pass@2 on ARC-AGI-1) with zero verbalized reasoning — direct evidence that latent test-time compute scales without chain-of-thought. Its Section 6.6 also supplies a second, equally important lesson: comparing MIN vs. STANDARD effort, all four reported endpoints favor STANDARD in point estimate, but the difference is not statistically significant (McNemar $p = 0.167$). Sometimes you cannot tell whether extra compute helped at all.

BDH (the base architecture) has a narrower, honestly-scoped role here: it has *no* depth dial of its own — its recurrence is over tokens with fixed layers (Eq. 8), and BDH-CQ's `H_r` reasoning loop is an addition on top, not something BDH itself provides. We state this explicitly rather than inflating BDH's relevance to inference-time scaling specifically.

Evidence discipline

Every claim in the interactive artifact carries an explicit tier: **orange** (our own sweep, checkable against gold answers), **green** (live, run just now, unverified since a user-typed question has no gold answer), **purple** (published by the original authors, not reproduced by us), or **grey** (architectural reasoning, no chart). Two cells are architecturally blocked, not merely unstudied: beam search and budget forcing both require a decoded, inspectable intermediate trace to intervene on, and BDH-CQ's `H_r` states are never decoded into language (BDH-CQ report, Section 1). This is the same root cause producing two independent blocked cells — not a coincidence.

Comparison to incumbents and limitations

Against standard fixed-depth transformer inference, all four methods trade latency and/or compute for accuracy on reasoning-heavy tasks; none is universally best. Best-of-N and budget forcing are architecture-agnostic and cheap to add to any autoregressive model. Latent-depth recurrence (Huginn) and BDH-CQ's dual recurrence require training the base model for variable-depth computation from the start — they cannot be bolted onto an existing fixed-depth model. BDH-CQ's checkpoint is not public and its ranker internals are proprietary, so its own best-of-N-style candidate construction (Section 3.1) cannot be independently swept, only cited.

The clearest limitation across all four methods, replicated in our own data: saturation is task-dependent, not universal. In the paper's own numbers, OpenBookQA accuracy under Huginn's recurrence gains +7.2pp (r=4→8), then +2.2pp (r=8→16), then +1.2pp (r=16→32) — visibly flattening — while GSM8K CoT keeps climbing hard over the same range (0% at r=1 to 34.8% at r=32). Same dial, same model, two different saturation points. Extrapolating from any single task's curve to "compute always helps" or "compute never helps past X" is the most common misconception this artifact is built to correct.

Primary sources

Geiping et al. 2025 (arXiv:2502.05171); Snell et al. 2024 (arXiv:2408.03314); Kosowski et al. 2025 (arXiv:2509.26507); Engdahl et al. 2026 (arXiv:2608.09888); Muennighoff et al. 2025 (arXiv:2501.19393). All equations and quotes attributed to these sources were verified character-by-character against the primary PDFs, not reproduced from memory.